

CH-512

Machine Learning for Process Engineering



Soft sensor development for debutanizer distillation column

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2025CHM1002

Basic steps for the data given :

- 1:-Cleaned data with 2394 rows saved to debutanizer_data_clean.xlsx
- 2:-Training data: 1675 rows saved to train_data.xlsx
- 3:-Testing data: 719 rows saved to test_data.xlsx
- 4:-Scaled training and test data saved as train_data_scaled.xlsx and test_data_scaled.xlsx

Training shapes: $X=(1675, 6)$, $y=(1675, 1)$
Testing shapes: $X=(719, 6)$, $y=(719, 1)$

Feature Selection Performed

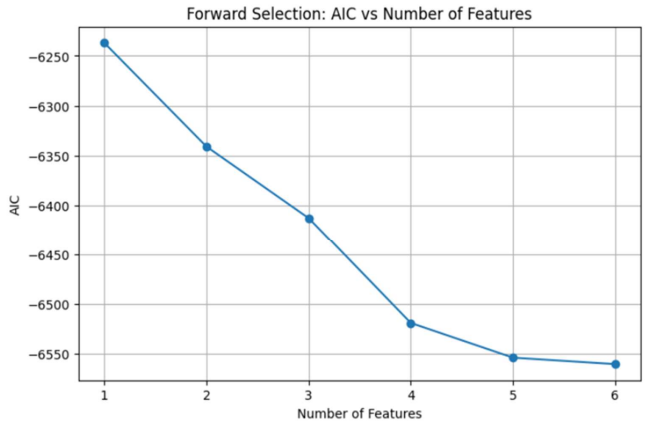
- ❌ Dropped due to high correlation: ['u7']
- ✅ Remaining features after correlation filtering: ['u1', 'u2', 'u3', 'u4', 'u5', 'u6']

🔍 Starting forward selection...

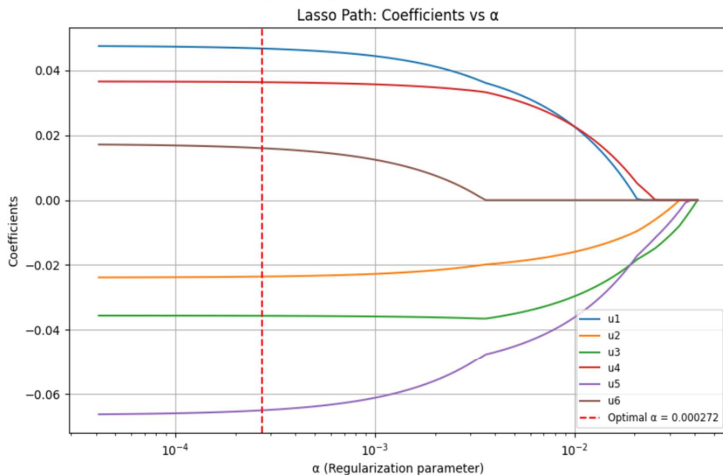
- + Added feature: u3, SSE improvement: inf
- + Added feature: u5, SSE improvement: 2.480771
- + Added feature: u1, SSE improvement: 1.636849
- + Added feature: u4, SSE improvement: 2.268433
- + Added feature: u2, SSE improvement: 0.739867
- + Added feature: u6, SSE improvement: 0.167707

Selected Features based on correlation: ['u1', 'u2', 'u3', 'u4', 'u5', 'u6']

Forward Selection

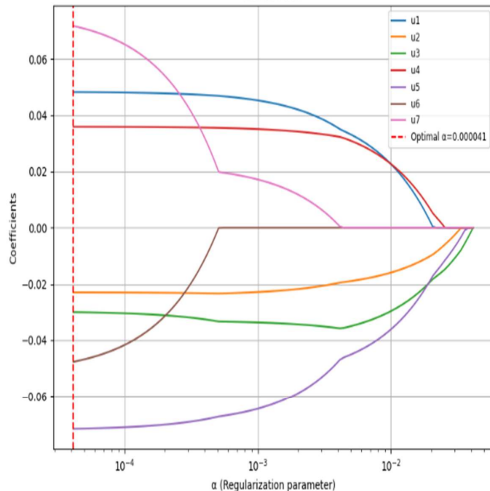


- ❌ Dropped due to high correlation: ['u7']
- ✅ Remaining features after correlation filtering: ['u1', 'u2', 'u3', 'u4', 'u5', 'u6']
- 🔍 Optimal α found by cross-validation: 0.000272
- 🏠 Features selected by Lasso Regression: ['u1', 'u2', 'u3', 'u4', 'u5', 'u6']

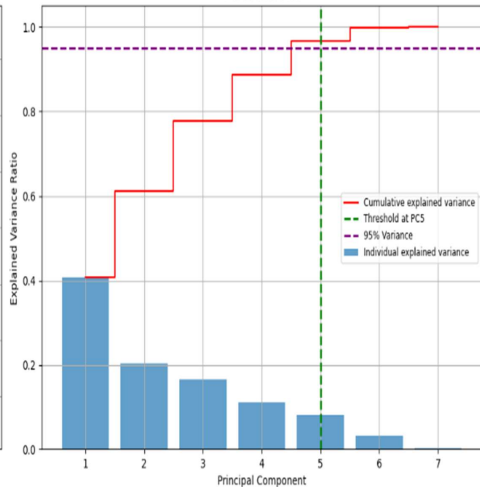


Cross validation =
0.000272

Lasso Path



PCA Scree Plot



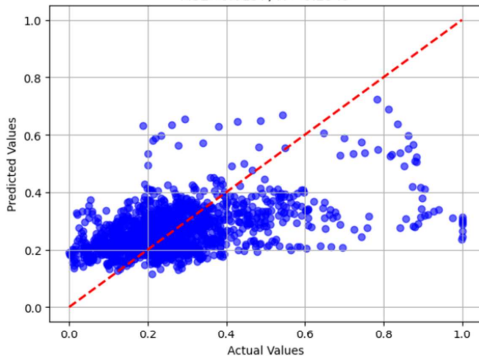
Number of principal components selected for 95% variance: 5

✓ Selected features (Forward Selection): ['u3', 'u5', 'u1', 'u4', 'u2', 'u6']

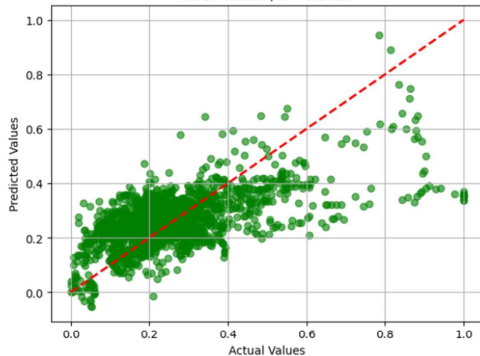
First-order model: $MSE = 0.0197$, $R^2 = 0.2349$

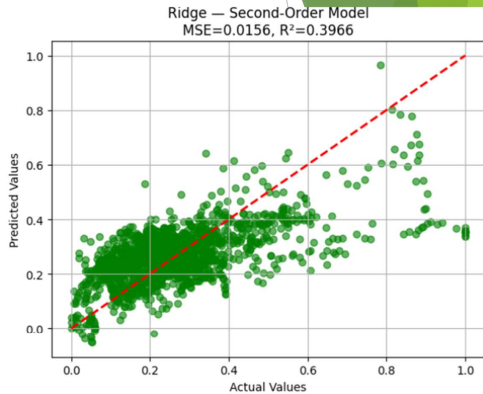
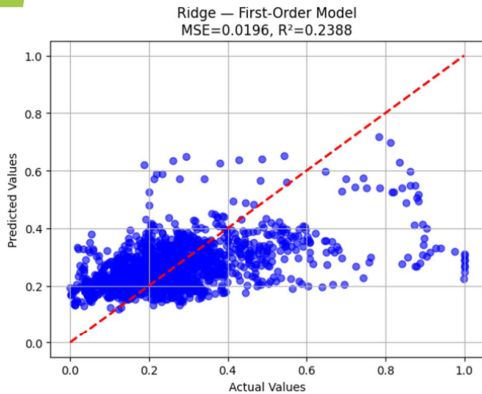
Second-order model: $MSE = 0.0159$, $R^2 = 0.3827$

First-Order Model
 $MSE=0.0197$, $R^2=0.2349$

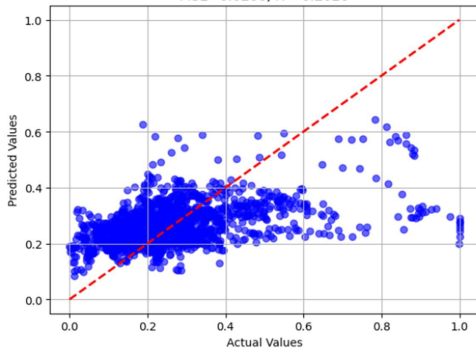


Second-Order Model
 $MSE=0.0159$, $R^2=0.3827$

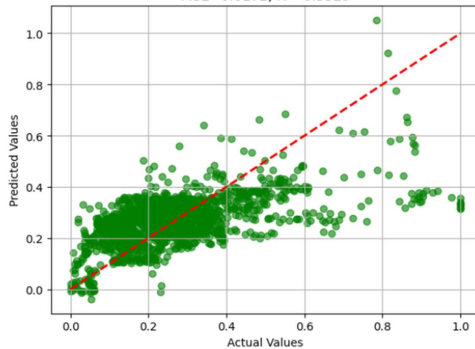




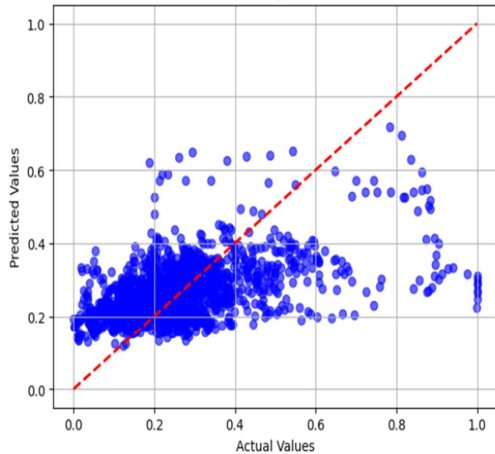
PCA Regression — First-Order
MSE=0.0206, $R^2=0.2026$



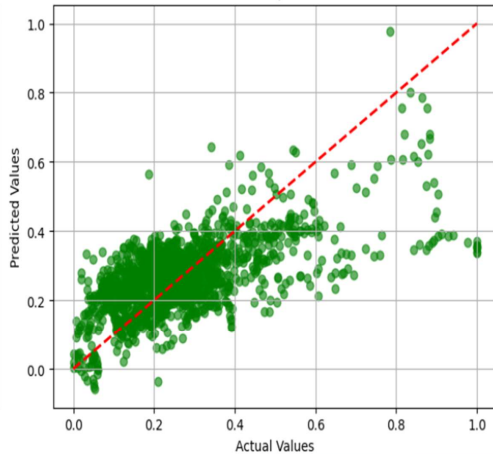
PCA Regression — Second-Order
MSE=0.0172, $R^2=0.3329$



Lasso — First-Order Model
 $\text{MSE}=0.0196$, $R^2=0.2387$



Lasso — Second-Order Model
 $\text{MSE}=0.0155$, $R^2=0.3980$



Parametric function approximation techniques:

Selected Features based on correlation: ['u1', 'u2', 'u3', 'u4', 'u5', 'u6']

Training Shallow NN...

Shallow NN -> SSE: 21.1300, R2: 0.0875

Training Deep NN...

Deep NN -> SSE: 21.3065, R2: 0.0799

Starting Hyperparameter Tuning...

Best Configuration: LR=0.01_Reg=0.001_L=[16] with SSE: 16.6453

AR Model Features: ['u1', 'u2', 'u3', 'u4', 'u5', 'u6', 'y_lag_1', 'y_lag_2']

Training Autoregressive NN...

Autoregressive NN -> SSE: 0.7328, R2: 0.9683

Training LSTM Model...

LSTM -> SSE: 28.0614, R2: -0.2118

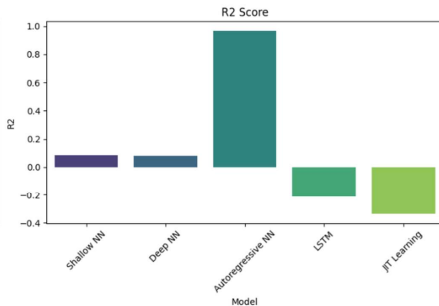
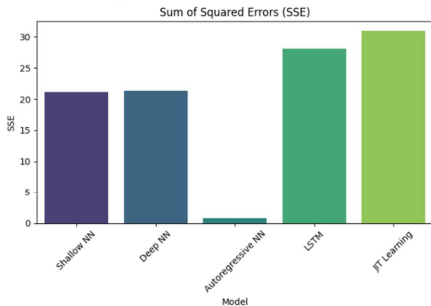
Non-Parametric function approximation techniques:

```
Starting JIT Learning on 719 test points...
This may take a few minutes. Using tqdm for progress tracking.
 0%|          | 2/719 [00:06<37:05, 3.10s/it]WARNING:tensorflow:
 0%|          | 3/719 [00:08<29:22, 2.46s/it]WARNING:tensorflow:
100%|██████████| 719/719 [19:58<00:00, 1.67s/it]
JIT Learning -> SSE: 30.9204, R2: -0.3353
```

Predictive Performance of all the Developed Models

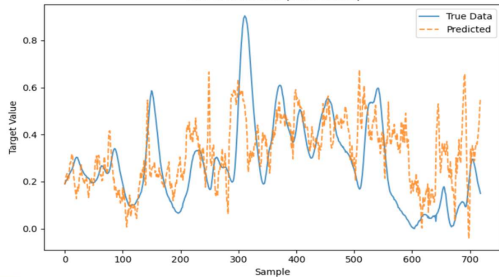
=== Final Model Comparison ===

	Model	SSE	R2
0	Shallow NN	21.130025	0.087520
1	Deep NN	21.306452	0.079901
2	Autoregressive NN	0.732830	0.968332
3	LSTM	28.061412	-0.211805
4	JIT Learning	30.920408	-0.335268

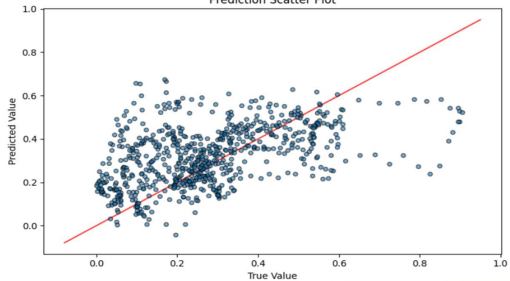


Analysis: Shallow NN

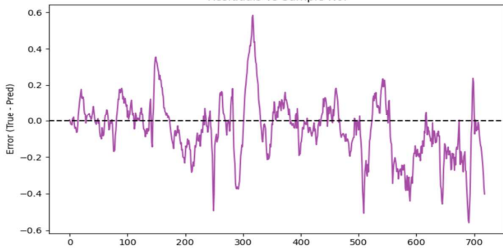
True vs Predicted (Time Series)



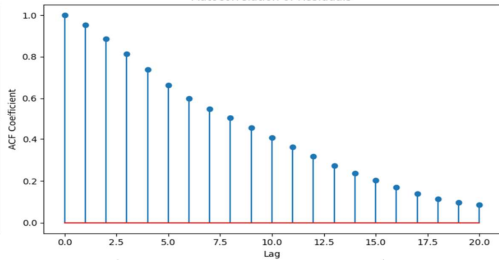
Prediction Scatter Plot



Residuals vs Sample No.

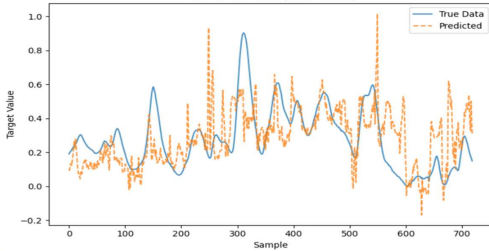


Autocorrelation of Residuals

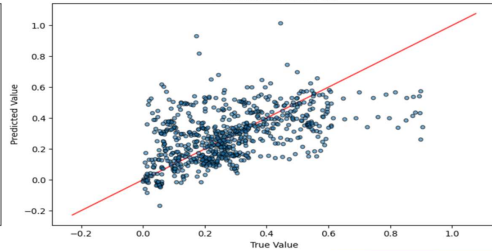


Analysis: Deep NN

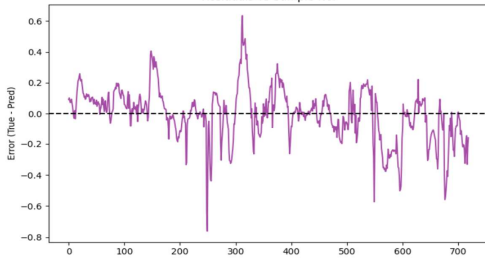
True vs Predicted (Time Series)



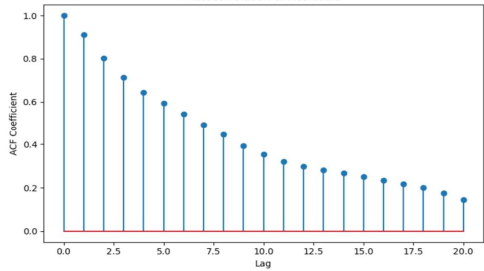
Prediction Scatter Plot



Residuals vs Sample No.

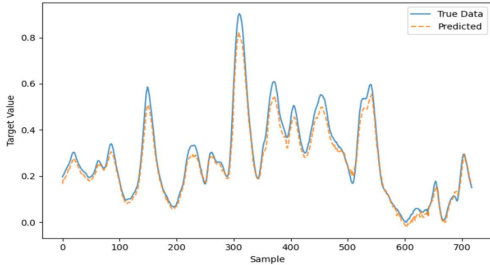


Autocorrelation of Residuals

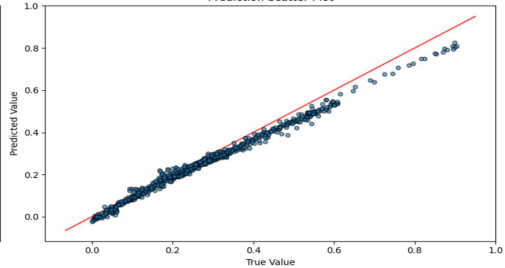


Analysis: Autoregressive NN

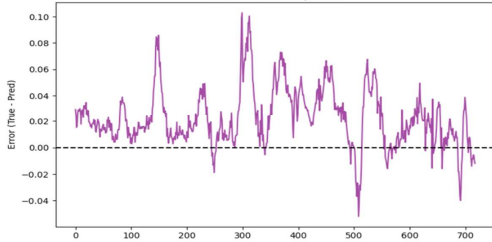
True vs Predicted (Time Series)



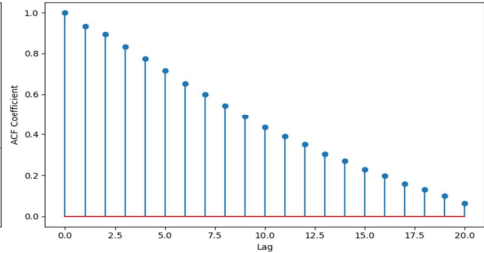
Prediction Scatter Plot



Residuals vs Sample No.

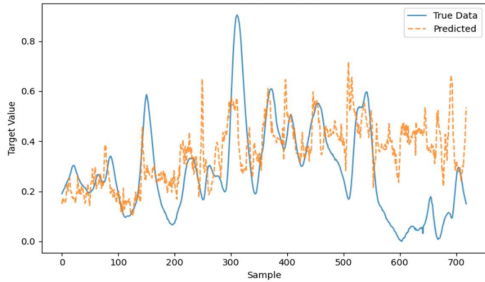


Autocorrelation of Residuals

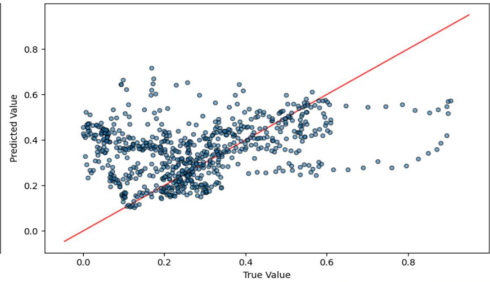


Analysis: LSTM

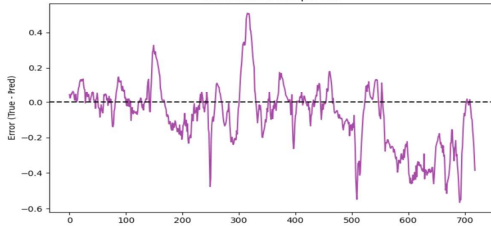
True vs Predicted (Time Series)



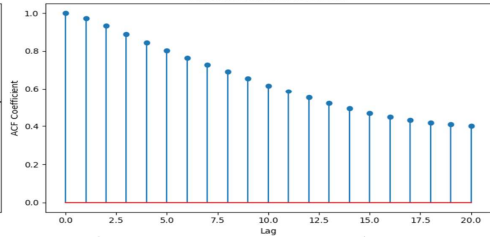
Prediction Scatter Plot



Residuals vs Sample No.

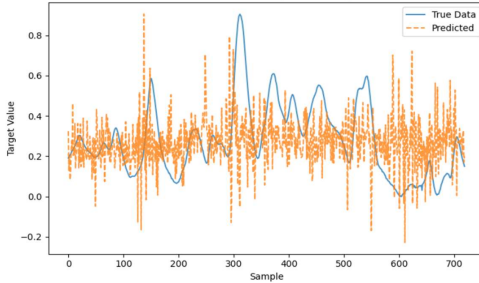


Autocorrelation of Residuals

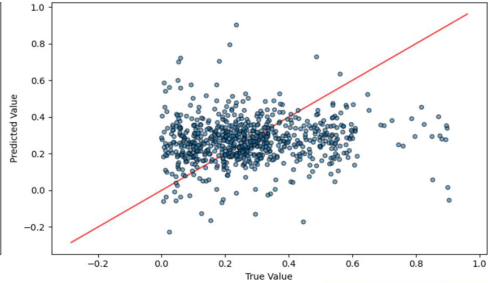


Analysis: JIT Learning

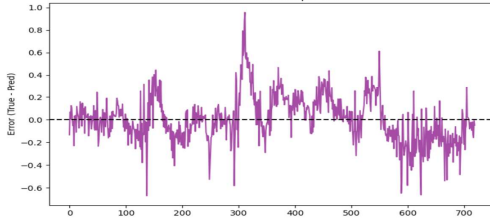
True vs Predicted (Time Series)



Prediction Scatter Plot



Residuals vs Sample No.



Autocorrelation of Residuals

